4COSC002W Mathematics for Computing

Lecture 9

Probability Theory

UNIVERSITY OF WESTMINSTER

What is Probability?

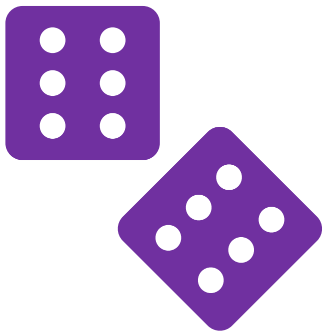
Probability measures the likelihood of an event occurring.

Output range: Between *0 (impossible)* and *1 (certain)* – [0,1]

For example:

The chance that a human lives for 1000 years is **0.**

The chance we all die someday is **1.**



Applications:

Critical in computing for decision-making, algorithms, and risk assessment.

Theoretical Probability

Probability of event A occurring:

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

Number of favourable outcomes: The count of outcomes that are considered as a success or the desired outcome for the event A.

Total number of possible outcomes (probability space): The total count of all possible outcomes for the event, assuming each outcome is equally likely.

NB! Theoretical probability assumes that every outcome has an *equal chance of occurring*.

Common Examples of Probability

Examples: Theoretical Probaility

Rolling a Die: The probability of *rolling a 4* on a six-sided die. There is one favorable outcome (rolling a 4), and six possible outcomes.

$$\mathrm{P}(4) = \frac{1}{6}$$

Flipping a Coin: The probability of *getting heads* when flipping a fair coin. There is one favorable outcome (heads), and two possible outcomes (heads or tails).

$$\mathrm{P}(\mathrm{Heads}) = \frac{1}{2}$$

Drawing a Card: The probability of *drawing an Ace* from a standard deck of 52 cards. There are four favorable outcomes (the four Aces), and 52 possible outcomes.

$$\mathrm{P}(\mathrm{Ace}) = \frac{4}{52} = \frac{1}{13}$$

Complementary Events

Events that together cover all possible outcomes are called *complementary*.

If $\boldsymbol{A}$ is an event, the *complement of* $\boldsymbol{A}$, denoted as $\boldsymbol{A'}$ or sometimes $\overline{\boldsymbol{A}}$, represents all outcomes that are *not in* $\boldsymbol{A}$. The probability of the complementary event $\boldsymbol{A'}$ is given by:

$$\boldsymbol{P(A') = 1 - P(A)}$$

$P(A)$ is the probability of event A occurring.
$P(A')$ is the probability of event A not occurring.

Example:
If the probability of raining is 0.3, the probability of not raining is 0.7.

Examples: Complementary Events

Coin Toss:

If $P(\text{Heads}) = 0.5$, then $P(\text{Tails}) = 1 - P(\text{Heads}) = 1 - 0.5 = 0.5$.

Drawing a Card:

If the probability of drawing a king from a standard deck of cards is $\frac{4}{52}$ or $\frac{1}{13}$, then the probability of not drawing a king is $1 - \frac{1}{13} = \frac{12}{13}$.

Dice Roll:

If the probability of rolling a 6 on a standard six-sided die is $\frac{1}{6}$, then the probability of not rolling a 6 is $1 - \frac{1}{6} = \frac{5}{6}$.

Independent Events

Independent events in probability theory are two or more events whose occurrence or non-occurrence does not affect the probability of the other events occurring.

If two events, A and B, are independent, the occurrence (or non-occurrence) of A does not change the probability of B occurring, and vice versa.

For two independent events, A and B, the probability of both events occurring is given by the product of their individual probabilities (*multiplication law*):

$$\mathbf{P(A \cap B) = P(A) \times P(B)}$$

Examples: Independent Events

Coin Tosses:

Tossing a coin twice. The outcome of the first toss (say, heads) does not affect the probability of the outcome of the second toss. If $P(\text{Heads}) = 0.5$ for each toss, then $P(\text{Heads and Heads}) = 0.5 \times 0.5 = 0.25$.

Dice and Cards:

Rolling a die and drawing a card from a deck. The probability of rolling a 6 (1/6) and drawing an ace (4/52) are independent events. So, $P(6 \text{ and Ace}) = \frac{1}{6} \times \frac{4}{52}$.

Traffic Lights:

The probability of hitting a green light on two different traffic signals might be independent if the lights are not synchronised. If the probability of green at each light is 0.3, then $P(\text{Green at both}) = 0.3 \times 0.3 = 0.09$.

The probability of rolling certain totals for a Pair of Dice

Total on Dice	Pairs of Dice	Probability
2	1+1	1/36 = 3%
3	1+2, 2+1	2/36 = 6%
4	1+3, 2+2, 3+1	3/36 = 8%
5	1+4, 2+3, 3+2, 4+1	4/36 = 11%
6	1+5, 2+4, 3+3, 4+2, 5+1	5/36 = 14%
7	1+6, 2+5, 3+4, 4+3, 5+2, 6+1	6/36 = 17%
8	2+6, 3+5, 4+4, 5+3, 6+2	5/36 = 14%
9	3+6, 4+5, 5+4, 6+3	4/36 = 11%
10	4+6, 5+5, 6+4	3/36 = 8%
11	5+6, 6+5	2/36 = 6%
12	6+6	1/36 = 3%

Probabilities of Outcomes in Three-Coin Toss

Specific Outcome	Probability of Specific Outcome	Total Heads in Outcome	Cumulative Probability of Total Heads (%)	Probability Calculation
TTT	1/8	0	12.5%	1/8 (1 outcome out of 8)
TTH	1/8	1	37.5%	3/8 (3 outcomes out of 8)
HTT	1/8	1	37.5%	3/8 (3 outcomes out of 8)
THT	1/8	1	37.5%	3/8 (3 outcomes out of 8)
THH	1/8	2	37.5%	3/8 (3 outcomes out of 8)
HHT	1/8	2	37.5%	3/8 (3 outcomes out of 8)
HTH	1/8	2	37.5%	3/8 (3 outcomes out of 8)
HHH	1/8	3	12.5%	1/8 (1 outcome out of 8)

Conditional Probability

Conditional probability measures the probability of an event occurring, given that another event has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0$$

Interpretation: It quantifies how the probability of one event changes when another related event is known to occur.

Example:
In a standard deck of 52 cards: *Event A*: Drawing an ace. *Event B*: Drawing a red card.

If $P(B) = \frac{26}{52}$ and $P(A \cap B) = \frac{2}{52}$, then: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{52}}{\frac{26}{52}} = \frac{2}{26} = \frac{1}{13}$.

Example: Conditional Probability

A fair die is rolled. Let **A** be the event that the outcome is an *odd number*, i.e., A={1,3,5}. Also let **B** be the event that the outcome is less than or equal to 3, i.e., B={1,2,3}.

- What is the Probability Space S?
- What is P(A)?
- What is P(B)?
- What is P(A∩B)?
- What is the conditional probability P(A|B)?

Example: Conditional Probability

Solution:

Probability Space:

$$S=\{1,2,3,4,5,6\}$$

Probability of A:

$$P(A)=\frac{|A|}{|S|}=\frac{|\{1,3,5\}|}{6}=\frac{3}{6}=\frac{1}{2}$$

Probability of B:

$$P(B)=\frac{|B|}{|S|}=\frac{|\{1,2,3\}|}{6}=\frac{3}{6}=\frac{1}{2}$$

Example: Conditional Probability

Given sets: A={1,3,5}   B={1,2,3}

Let's find P(A∩B):

First, we need to define A∩B={1,3} – 2 common favourable outcomes.

$$P(A\cap B)=\frac{|A\cap B|}{|S|}=\frac{|\{1,3\}|}{6}=\frac{2}{6}=\frac{1}{3}$$

Example: Conditional Probability

Now we can find the conditional probability of A given that B occurred as we know:

$$P(A \cap B) = \frac{1}{3} \text{ and } P(B) = \frac{1}{2}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{1}{3} \text{ x } \frac{2}{1} = \frac{2}{3}.$$

Bayes' Theorem

Bayes' Theorem is an extension of conditional probability:

$$\mathbf{P(A|B)} = \frac{\mathbf{P(B|A)} \times \mathbf{P(A)}}{\mathbf{P(B)}}$$

where:
$P(A|B)$: The probability of event A occurring given that B is true.
$P(B|A)$: The probability of event B occurring given that A is true.
$P(A)$: The prior or initial probability of event A.
$P(B)$: The prior or initial probability of event B.

NB! Bayes' Theorem Formula is useful when you know P(B|A), P(A), and P(B) but need to calculate P(A|B).

Bayes' Theorem

Bayes' Theorem is a cornerstone of probability theory, allowing us to update beliefs based on new evidence.

Key Applications:

- Medical diagnostics
- Spam email filtering
- Machine learning (e.g., Naive Bayes classifier)

Example:
Spam Filtering with Bayes' Theorem

Scenario:

An email filter detects spam with 95% accuracy. Only 5% of all emails are spam. The filter also incorrectly flags 1% of non-spam emails as spam.

If the filter marks an email as spam, *what is the chance that it is actually spam*?

Using Bayes' Theorem:

$$P(SPAM|FILTER) = \frac{P(FILTER|SPAM) \times P(SPAM)}{P(FILTER)}$$, where:

$P(SPAM|FILTER)$: Probability the email is spam, given it was marked as spam.

$P(FILTER|SPAM) = 0.95$: The probability the filter correctly identifies spam.

$P(SPAM) = 0.05$: Probability of any email being spam.

$P(FILTER)$: Probability the filter marks an email as spam.

Example:
Spam Filtering with Bayes' Theorem

STEP 1: CALCULATE P(FILTER):

<u>What is P(FILTER)?</u> It's the chance that the filter marks any email as spam (in general).
This includes two possibilities:

- The email is *actually spam*, and the filter correctly identifies it.
- The email is *not spam*, but the filter incorrectly marks it as spam (a false positive).

Thus:

- Correct spam detections ($FILTER \cap SPAM$): $P(FILTER|SPAM) \times P(SPAM) = 0.95 \times 0.05 = 0.0475$.
- False positives ($FILTER \cap NOT\ SPAM$): $P(FILTER|NOT\ SPAM) \times P(NOT\ SPAM) = 0.01 \times (1 - 0.05) = 0.0095$.

Adding these together gives: $P(FILTER) = 0.0475 + 0.0095 = \mathbf{0.057}$.

Example:
Spam Filtering with Bayes' Theorem

STEP 2: APPLY BAYES' THEOREM:

Now substitute into the formula for $P(SPAM|FILTER)$:

$$P(SPAM|FILTER) = \frac{P(FILTER|SPAM) \times P(SPAM)}{P(FILTER)} = \frac{0.95 \times 0.05}{0.057} = \frac{0.0475}{0.057} \approx 0.8333.$$

The probability that an email marked as spam is actually spam is approximately **83.33%**.

Key Takeaway:

Even though the filter is very accurate (95%), its effectiveness also depends on how often spam occurs and the rate of false positives. In this case, the rarity of false positives (1%) means the filter performs well, giving a high probability (83.33%) that a flagged email is truly spam.